

Assignment 10

$$1. \quad \lambda = \frac{60}{60} = 1, \quad t = 1$$

$$(i) \quad P[X(1) = 2] = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!} = \frac{e^{-1} \cdot (1)^2}{2!} = \boxed{0.1839}$$

$$(ii) \quad P[X(1) < 2] = P[X(1) = 0] + P[X(1) = 1] + P[X(1) = 2]$$

$$= \frac{e^{-1} \cdot 1}{1!} + \frac{e^{-1} \cdot 1}{1!} + 0.1839$$

$$= e^{-1} + e^{-1} + 0.1839$$

$$= \boxed{0.9197}$$

$$2. \quad \lambda = \frac{30}{60} = 0.5, \quad t = 3$$

$$(i) \quad P[X(3) = 0] = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!} = \frac{e^{-(0.5 \times 3)} \times (0.5 \times 3)^0}{0!}$$

$$= \boxed{0.2231}$$

$$(ii) \quad t = 5, \quad \lambda t = 2.5$$

$$P[X(5) > 4] = 1 - [P[X(5) = 0] + P[X(5) = 1] + P[X(5) = 2] + P[X(5) = 3] + P[X(5) = 4]]$$

$$= 1 - \left[\frac{e^{-2.5}}{0!} + \frac{e^{-2.5} \times 2.5}{1!} + \frac{e^{-2.5} \cdot (2.5)^2}{2!} + \frac{e^{-2.5} \cdot (2.5)^3}{3!} + \frac{e^{-2.5} \cdot (2.5)^4}{4!} \right]$$

$$= \boxed{0.1088}$$

3. $\lambda = 2, t = 5$

$$P[X(5) = 0] = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!} = \frac{e^{-10} \cdot (10)^0}{0!} = \boxed{4.54 \times 10^{-5}}$$

4. $\lambda = 3, t = 2$

(i) $P[X(2) = 4] = \frac{e^{-\lambda t} (\lambda t)^x}{x!} = \frac{e^{-6} \cdot (6)^4}{4!} = \boxed{0.1338}$

(ii) $P[X(2) \geq 4] = 1 - [P[X(2) = 0] + P[X(2) = 1] + P[X(2) = 2] + P[X(2) = 3]]$
 $= \boxed{0.8488}$

5. $\lambda = 3, t = 1$

(i) $P(t > 1) = ?$

$$f(t) = \lambda e^{-\lambda t} = 3e^{-3t}$$

$$P(t > 1) = \int_1^{\infty} 3e^{-3t} dt = 3 \left[\frac{e^{-3t}}{-3} \right]_1^{\infty} = \left[-e^{-3t} \right]_1^{\infty}$$

$$= 0 + e^{-3}$$

$$= \boxed{0.0498}$$

(ii) $P(1 < t < 2) = \int_1^2 3e^{-3t} dt = \left[-e^{-3t} \right]_1^2$
 $= -e^{-6} + e^{-3} = \boxed{0.0473}$

(iii) $P(t \leq 3) = 1 - \int_3^{\infty} 3e^{-3t} dt$

$$= 1 + \left[\frac{e^{-3t}}{-3} \right]_3^{\infty} = 1 + [0 - e^{-9}]$$

$$= \boxed{0.9998}$$

6. $\lambda = 6, p = 0.5$

$$\lambda_{\text{new}} = \lambda_{\text{old}} \times p = 6 \times 0.5 = 3$$

$$t = 4$$

$$\lambda t = 12$$

$$P[X(4) = 10] = \frac{e^{-12} \cdot (12)^{10}}{10!} = 0.1048$$

7. $\lambda = 6$

(i) $t = 1$

$$\lambda t = 6$$

$$P[X(1) = 0] = \frac{e^{-6} \cdot (6)^0}{0!} = 0.002478$$

(ii) $P[X(1) = 6] = \frac{e^{-6} \cdot (6)^6}{6!} = 0.1606$

(iii) $t = 0.5 \rightarrow \lambda t = 3$

$$P[X(0.5) < 3] = P[X(0.5) = 0] + P[X(0.5) = 1] + P[X(0.5) = 2]$$

$$= \frac{e^{-3} \cdot (3)^0}{0!} + \frac{e^{-3} \cdot (3)^1}{1!} + \frac{e^{-3} \cdot (3)^2}{2!}$$

$$= 0.4232$$

10. $\lambda = 2$

(i) $t = 0.5$

$$\lambda t = 1$$

$$P[X(0.5) = 1] = \frac{e^{-1} \cdot (1)^1}{1!} = 0.3678$$

(ii) $t = 2$

$$\lambda t = 4$$

$$P[X(2) = 3] = \frac{e^{-4} \cdot (4)^3}{3!} = 0.1953$$

8. $\lambda = 1/5 = 0.2$ per week

$t = 7$

$$P[X(t) = 3] = 1 - [P[X(7) = 0] + P[X(7) = 1] + P[X(7) = 2]]$$

$$= \boxed{0.1665}$$

probability that all chips fail in first 7 weeks

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9. $\lambda = 1$

(i) $P(t > 2) = \int_2^{\infty} \lambda e^{-\lambda t} = \int_2^{\infty} e^{-t} = [-e^{-t}]_2^{\infty} = \boxed{0.1353}$

(ii) $P[X(10) \leq 5] = P[X(10) = 0] + P[X(10) = 1] + P[X(10) = 2] + P[X(10) = 3] + P[X(10) = 4] + P[X(10) = 5]$

$$= \boxed{0.067}$$

RVP
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